Fiscal Year:	FY 2024	**Task Last Updated:**	FY 07/12/2024
PI Name:	Weaver, Ashley Ph.D.		
Project Title:	Quantitative CT and MRI-based Modeling Assessment of Dynamic Vertebral Strength and Injury Risk Following Long-Duration Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **Dynamic Loads**:Risk of In-Mission Injury and Performance Decrements and Long-term Health Effects due to Dynamic Loads		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	336-716-0944
Organization Name:	Wake Forest University		
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PI Web Page:			
City:	Winston-Salem	**State:**	NC
Zip Code:	27101-4101	**Congressional District:**	12
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015-16 HERO NNJ15ZSA001N-Crew Health (FLAGSHIP, NSBRI, OMNIBUS). Appendix A-Crew Health, Appendix B-NSBRI, Appendix C-Omnibus
Start Date:	10/01/2016	**End Date:**	08/31/2025
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	4	**No. of Master' Degrees:**	1
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 8/31/2025 per NSSC information (Ed., 8/1/24) NOTE: End date changed to 8/31/2024 per S. Mack-Phillips/JSC (Ed., 7/18/23) NOTE: End date changed to 8/31/2023 per V. Lehman/JSC (Ed., 3/29/23) NOTE: End date changed to 8/31/2022 per L. Barnes-Moten/JSC (Ed., 8/2/21) NOTE: End date changed to 8/31/2021 per NSSC information/S. Huppman/HRP (Ed., 2/25/2020) NOTE: End date changed to 2/1/2020 per NSSC information (Ed., 7/8/19)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Stitzel, Joel Ph.D. (Wake Forest University) Tooze, Janet Ph.D. (Wake Forest University)		

Grant/Contract No.:	NNX16AP89G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Prolonged periods of near weightlessness can cause damage to astronauts' musculoskeletal system. This damage can increase the risk of skeletal tissue failure (e.g., fractures, tears) when experiencing forceful, dynamic loads. Fractures of the spine during dynamic conditions such as launch or landing could cause a mission to fail. This study will measure this degradation of astronauts' vertebrae and spinal muscles during missions aboard the International Space Station (ISS). We will then determine the extent of vertebral weakening of crewmembers during long-duration missions. Changes in pre- and post-flight vertebral geometry, volume, cortex thickness, and bone mineral density will be measured from existing lumbar quantitative computed tomography (qCT) scans, as well as from planned qCT scans of the cervical, thoracic, and lumbar spine from nine ISS crewmembers. Likewise, the pre- and post-flight spinal muscle volumes will be analyzed using both existing magnetic resonance imaging (MRI) scans and planned MRI scans from nine ISS crewmembers. The qCT and MRI scans will be analyzed to determine structural and material changes in the cervical, thoracic, and lumbar vertebrae and the spinal muscles that indicate damage which could weaken these tissues. Our unique engineering approach will measure the loss of vertebral strength during spaceflight conditions and predict the risk of failure during traumatic, dynamic loading conditions such as launch or landing. Vertebral strength and risk for vertebral fracture and injury will be quantified in dynamic simulations using a full human body model that is constructed using structural and material data gathered from the pre- and post-flight medical images for each astronaut. This study has significance in quantifying and addressing risks of long-duration spaceflight, including vertebral injury from dynamic loads, vertebral fracture, early onset vertebral osteoporosis due to spaceflight, and impaired performance due to reduced spinal muscle mass, strength, and endurance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Microgravity induces similar spinal changes to those seen in the aging population and people with limited mobility. Demonstrating how the vertebral column changes in response to microgravity can aid in refining the diagnostic and treatment protocols of physicians on Earth. Additionally, assessing vertebral column strength using finite element modeling can provide future techniques for assessing the efficacy of osteoporosis treatments, which would particularly benefit older adults.
Task Progress:	Prospective quantitative computed tomography (qCT) and magnetic resonance imaging (MRI) data collection has been completed for all nine of the enrolled subjects. Bone and muscle morphology and quality measures were completed. Different morphing methodologies to develop subject-specific finite element vertebrae models from the prospective CT data have been compared and optimum morphing methods for all the vertebrae have been identified. Pre- and post-flight subject-specific finite element vertebrae models are being developed using these methods. Vertebral compression test simulations are being conducted to assess the effects of space-induced changes on vertebral compression strength using these subject-specific vertebrae models. To study the effects of musculoskeletal changes on astronaut spinal injury risk, a modular deformable spine finite element human body model was developed by incorporating a deformable spine in the existing simplified human body model. This newly developed model has been validated against PMHS and volunteer test data from the literature to assess its biofidelity. Morphing has been used to create subject-specific versions of this model that match data derived from the astronaut's imaging. Work under student augmentation awards includes human body modeling of lunar rover scenarios and modeling lunar launch/landing events with small, medium, and large sizes of a human body model.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
Abstracts for Journals and Proceedings	Jeong T, Lalwala M, Devane K, Kauffman K, Gayzik FS, Stitzel JD, Weaver AA. "Development of subject-specific models to investigate effects of long-duration spaceflight on spine injury risk." 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. Abstracts. 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. , Feb-2023
Abstracts for Journals and Proceedings	Lalwala M, Devane K, Koya B, Gayzik FS, Weaver AA. "Spaceflight loading and muscle deconditioning effects on astronaut response." 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. Abstracts. 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. , Feb-2023
Abstracts for Journals and Proceedings	Lalwala M, Devane K, Koya B, Gayzik FS, Stitzel JD, Weaver AA. "Development of subject-specific models to investigate effects of long-duration spaceflight on spine injury risk." 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. Abstracts. 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. , Feb-2023
Abstracts for Journals and Proceedings	Poveda L, Lalwala M, Devane K, Gayzik FS, Stitzel JD, Weaver AA. "Setup and validation of simulated lunar terrain vehicle extra vehicular activities." 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. Abstracts. 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. , Feb-2023

Articles in Peer-reviewed Journals	Lalwala M, Devane KS, Koya B, Hsu FC, Yates KM, Newby NJ, Somers JT, Gayzik FS, Stitzel JD, Weaver AA. "Effect of active muscles on astronaut kinematics and injury risk for piloted lunar landing and launch while standing." Ann Biomed Eng. 2023 Jul;51(7):1408-1419. https://doi.org/10.1007/s10439-023-03143-y ; PMID: 36652027 , Jul-2023
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Articles in Peer-reviewed Journals	Poveda L, Devane K, Lalwala M, Gayzik FS, Stitzel JD, Weaver AA. "Injury risk predictions in Lunar Terrain Vehicle (LTV) Extravehicular Activities (EVAs): A pilot study. " Ann Biomed Eng. 2024 Sep;52(9):2534-45. Epub 2024 Jun 5. https://doi.org/10.1007/s10439-024-03543-8 ; PMID: 38836980; PMCID: PMC11329544. , Sep-2024